Using Detention Ponds

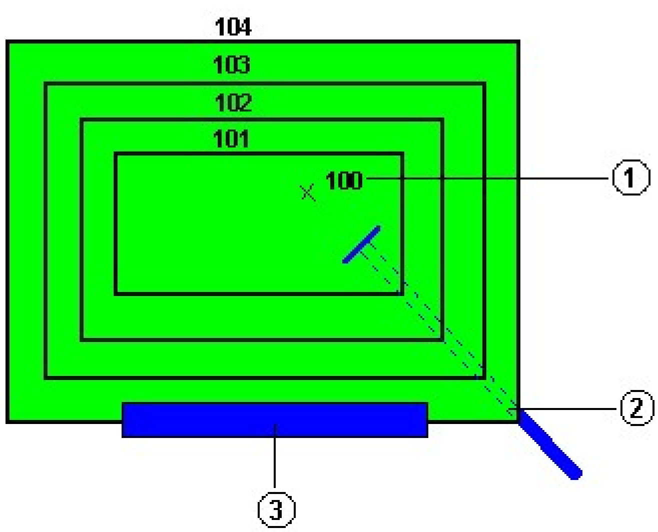
You can use a detention pond in pond routing procedures. After you set up the detention pond, Hydraflow Hydrographs Extension allows you to route any hydrograph through it. Before performing detention pond routing procedures, you must set up the stage-storage-discharge relationship, which governs detention ponds. The stage-storage-discharge relationship is represented by a table of values that determine how the pond performs when the water level reaches certain pre-determined depths or stages, similar to a pump performance curve (Q vs. head). In this situation, Q is paired with stage.

A typical stage-storage-discharge table could look as follows:

Stage (ft)	Storage (cuft)	Discharge (cfs)
0.00	0.00	0.00
1.00	5,400	5.75
2.00	11,780	11.89
3.00	17,000	21.43

For example, at a depth of two feet, this pond would produce an outflow of 11.89 cubic feet per second, while storing 11,780 cubic feet of water.

The following image shows a plan view of a typical detention pond.



(1) Elevation at stage 0	(2) Culvert structure	(3) Weir structure

Topics in this section

- Creating Detention Ponds
- Combining Detention Ponds
- Storage Tab - Stage/Storage/Discharge Dialog Box

  Use this tab to enter storage-related data when you set up detention ponds.
- Outlets Tab - Stage/Storage/Discharge Dialog Box

  Use this tab to enter the stage-storage-discharge data to specify the discharge values that correspond to the stage values.
- Pond Tools Tab - Stage/Storage/Discharge Dialog Box

  Use this tab to add and modify outlet structures and perform the routing of the detention ponds.
- Graphs Tab - Stage/Storage/Discharge Dialog Box

  Use this tab to view the plotted graphs.
- Table Tab - Stage/Storage/Discharge Dialog Box

  Use this tab to view the stage-storage-discharge data in numerical format.